MATH 201 Your name: _____

Learning target 1.6, version 1

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ (noun) is _____ (adjective), the _____ (noun) is _____ (adjective).

For each of the sentences they generated below, explain why it is correct or incorrect.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

(a) Whenever the **original function** is **decreasing**, the **derivative** is **negative**.

(b) Whenever the **derivative** is **increasing**, the **original function** is **positive**.

(c) Whenever the **second derivative** is **positive**, the **original function** is **concave up**.

(d) Whenever the **derivative** is **increasing**, the **second derivative** is **positive**.

MATH 201 Your name: _____

Learning target 1.6, version 2

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ (noun) is _____ (adjective), the _____ (noun) is _____ (adjective).

Unfortunately, class time ended before they finished a few of their sentences. Help them fill in the blanks correctly, and explain why your choices are correct.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

(a) Whenever the **derivative** is _____, the **second derivative** is **negative**.

(b) Whenever the **original function** is **concave up**, the **derivative** is _____.

(c) Whenever the _____ is **positive**, the _____ is **increasing**.

(d) Whenever the _____ is **negative**, the _____ is _____.
(There are several correct fill-ins!)

MATH 201 Your name: _____

Learning target 1.6, version 3

Explain the relationships between a function, its derivative, and its second derivative. Interpret information about these objects in real-world contexts.

Students in a recent calculus class were playing a Mad Libs game about the relationships between shapes of functions and values of the derivative:

Whenever the _____ (noun) is _____ (adjective), the _____ (noun) is _____ (adjective).

Unfortunately, class time ended before they finished a few of their sentences. Help them fill in the blanks correctly, and explain why your choices are correct.
(Hint: I don't think it's possible to generate a good explanation without saying the word "slope.")

(a) Whenever the **second derivative** is _____, the **derivative** is **increasing**.

(b) Whenever the **original function** is **concave down**, the **derivative** is _____.

(c) Whenever the _____ is **negative**, the _____ is **decreasing**.

(d) Whenever the _____ is **positive**, the _____ is _____.
(There are several correct fill-ins!)